MAT 201 - MATHEMATICAL METHODS I

LECTURE 4: MEAN VALUE THEOREM

LECTURE C24-1-4 :: WED., JAN. 08, 2025 2:30 PM (WAT) - SAT., JAN. 11, 2025 6:00 PM (WAT)

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LESSON OUTLINE

- Introduction
- Review: Limit, Continuity, Differentiability
- Rolle's Theorem
- Mean Value Theorem
- Summary

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INTRODUCTION

- The mean value theorem tells us (roughly) that if we know the slope of the secant line of a function whose derivative is continuous, then there must be a tangent line nearby with that same slope.
- We denote by $\mathbb{R}$ the set of real numbers. A domain *D* of $\mathbb{R}$ is any subset of $\mathbb{R}$.
- Typically this will be on open interval *(a,b)* or a closed interval *[a,b]*. A function of a real variable is a function $f : D \to \mathbb{R}$, where *D* is a domain of $\mathbb{R}$.

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REVIEW: LIMIT

Definition 1: We say that a function $f(x)$ has a **limit** L as x approaches a point x_0 and write
$\lim_{x \to x_0} f(x) = L,$
if for any $\epsilon > 0$ there exists $\delta > 0$ such that whenever $0 < |x - x_0| < \delta$ (and $x \in D$) we have $|f(x) - L| < \epsilon$.
The meaning of the above definition is that by choosing a sufficiently small interval $(x_0 - \delta, x_0 + \delta)$ of the point x_0 we can ensure that the values of $f(x)$ on this interval (excluding x_0) do not deviate from L by more than ϵ.

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REVIEW: CONTINUITY

Definition 2: We say that a function $f : D \to \mathbb{R}$ is **continuous** at a point $x_0 \in D$ if $\lim_{x \to x_0} f(x) = f(x_0)$.

Using the $\epsilon - \delta$ definition this can be stated as follows: given $\epsilon > 0$, there exists $\delta > 0$ such that whenever $|x - x_0| < \delta$ we have $|f(x) - f(x_0)| < \epsilon$.

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REVIEW: DIFFERENTIABILITY

Definition 3: Let $f(x)$ be defined on an interval $D \subset \mathbb{R}$. Let $x_0 \in D$. We say that $f(x)$ is differentiable at x_0 if the limit

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists.

The value of the limit is defined to be $f'(x_0)$, the derivative of f at x_0.

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ROLLE'S THEOREM

Definition 4: A point c is called a critical point of a differentiable function $f(x)$ if $f'(c) = 0$.

Rolle's Theorem: Suppose $f(x)$ is continuous on the interval $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$. Then there exists a number $c \in (a, b)$ such that $f'(c) = 0$.

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MEAN VALUE THEOREM

Suppose that $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists a point $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

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